

# Assignment 2 (AM5920): Kanak Agarwal (ID25M802)

## Question 1

Consider the wave propagation equation in a rod, given by,

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial x} \left( E(\epsilon) \frac{\partial u}{\partial x} \right)$$

with initial and boundary conditions,

$$u(x, 0) = u_0(x), \quad \frac{\partial u}{\partial t}(x, 0) = v_0(x)$$

and,

$$u(0, t) = 0, \quad u(L, t) = 0$$

where  $u(x, t)$  is the displacement at position  $x$  and time  $t$ ,  $E(\epsilon)$  is the Young's modulus which is a nonlinear function of the strain  $\epsilon(x, t) = \frac{\partial u}{\partial x}$ , and  $L$  is the length of the rod.

The displacement field is measured discretely at  $N_x$  spatial locations and  $N_t$  time steps, resulting in a dataset  $\{u(x_i, t_j)\}$  for  $i = 1, 2, \dots, N_x$  and  $j = 1, 2, \dots, N_t$ .

To ensure sufficient excitation of the nonlinear constitutive relation  $E(\epsilon)$ , the initial displacement and velocity fields are prescribed using a truncated Fourier sine series that satisfies the boundary conditions  $u(0, t) = u(L, t) = 0$ :

$$u_0(x) = \sum_{k=1}^K A_k \sin\left(\frac{k\pi x}{L}\right), \quad v_0(x) = \sum_{k=1}^K B_k \sin\left(\frac{k\pi x}{L}\right)$$

where  $K$  is the number of retained modes, which is chosen to be 5 in the current implementation.

The coefficients  $A_k$  and  $B_k$  are selected to provide broadband excitation and to ensure that the resulting strain field spans a wide range of values. This is important for accurately identifying the nonlinear Young's modulus  $E(\epsilon)$ .

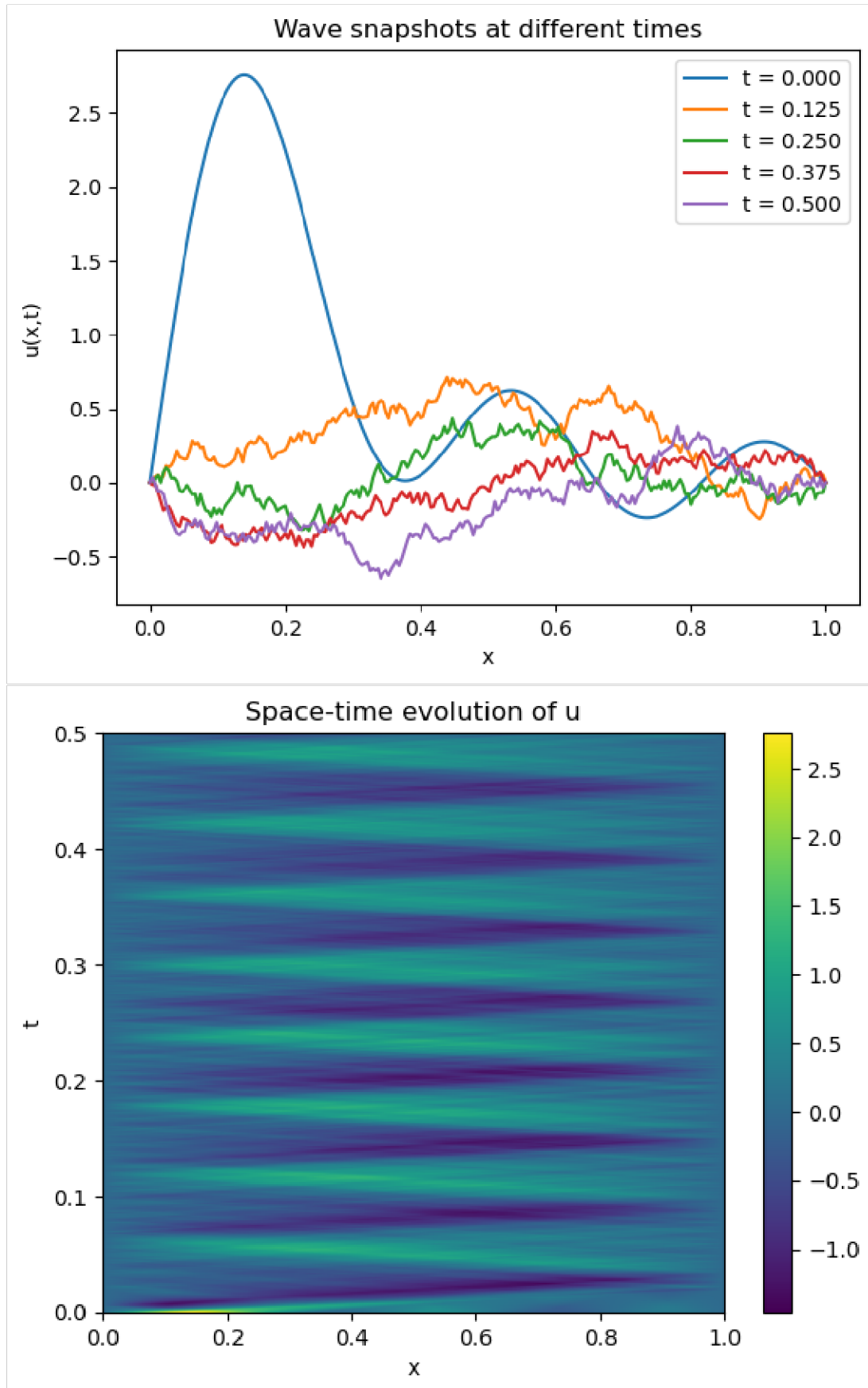
A practical choice is to sample the amplitudes from uniform distributions,

$$A_k \sim \mathcal{U}(0.3, 1.0), \quad B_k \sim \mathcal{U}(-0.5, 0.5)$$

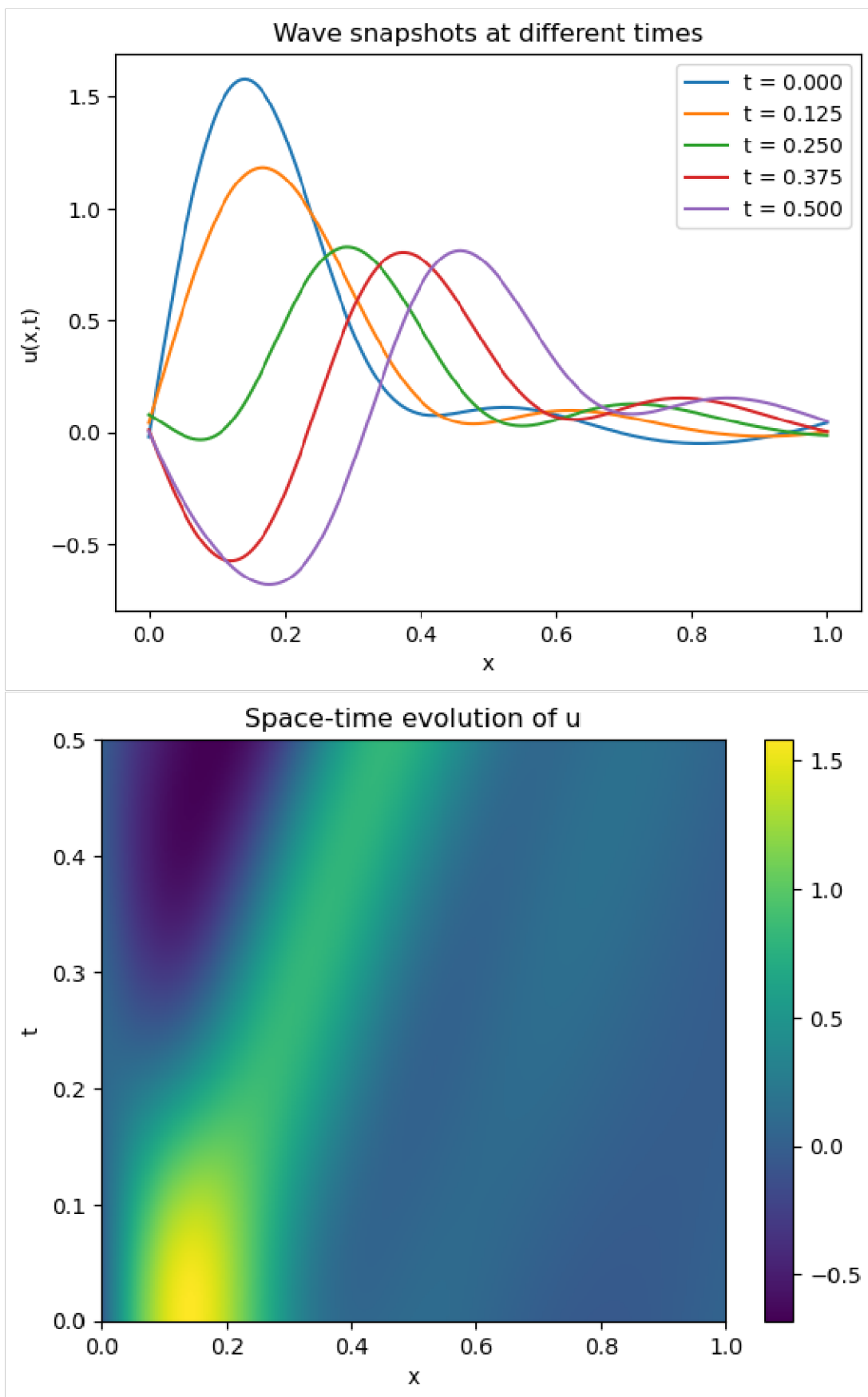
Some remarks on this choice of initial and boundary conditions,

- The sine basis ensures compatibility with Dirichlet boundary conditions.
- Including multiple modes introduces spatial variability and nonlinear interactions.
- Randomized amplitudes improve the identifiability of  $E(\epsilon)$  by exposing the system to a wider range of strain values.

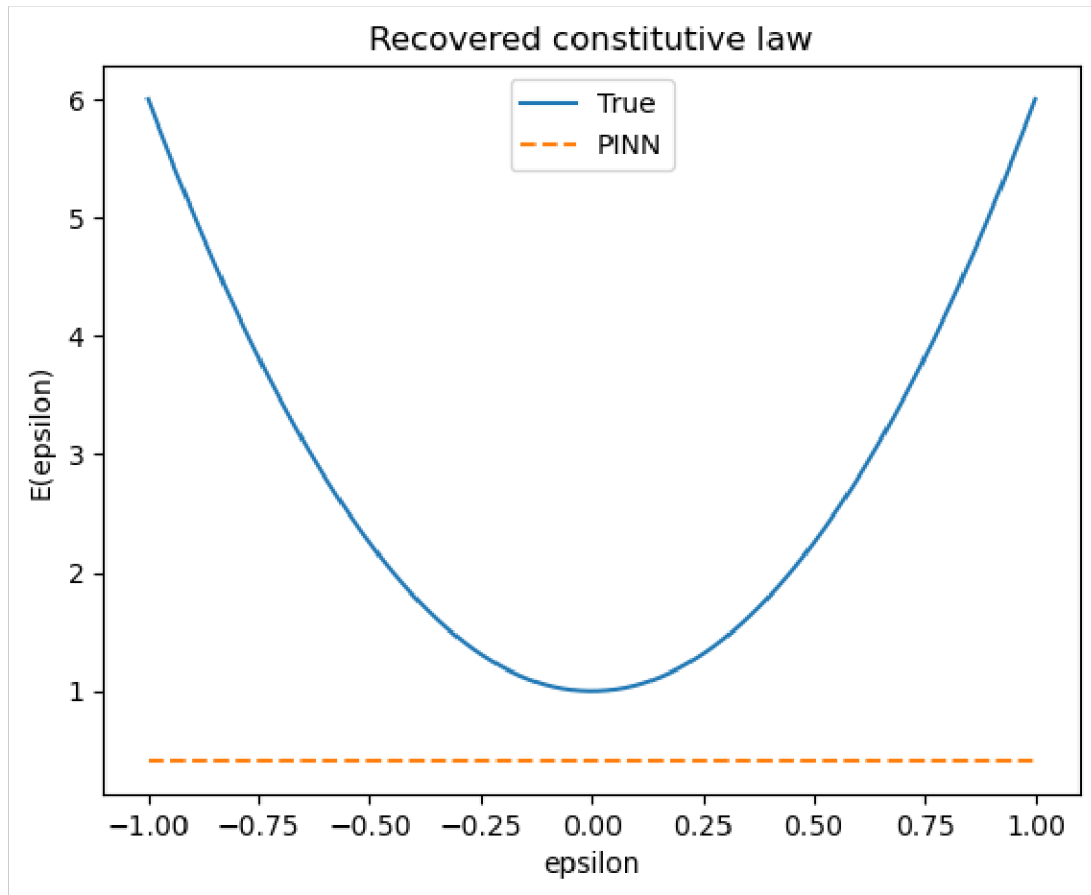
## Finite Difference Implementation



## PINN Implementation

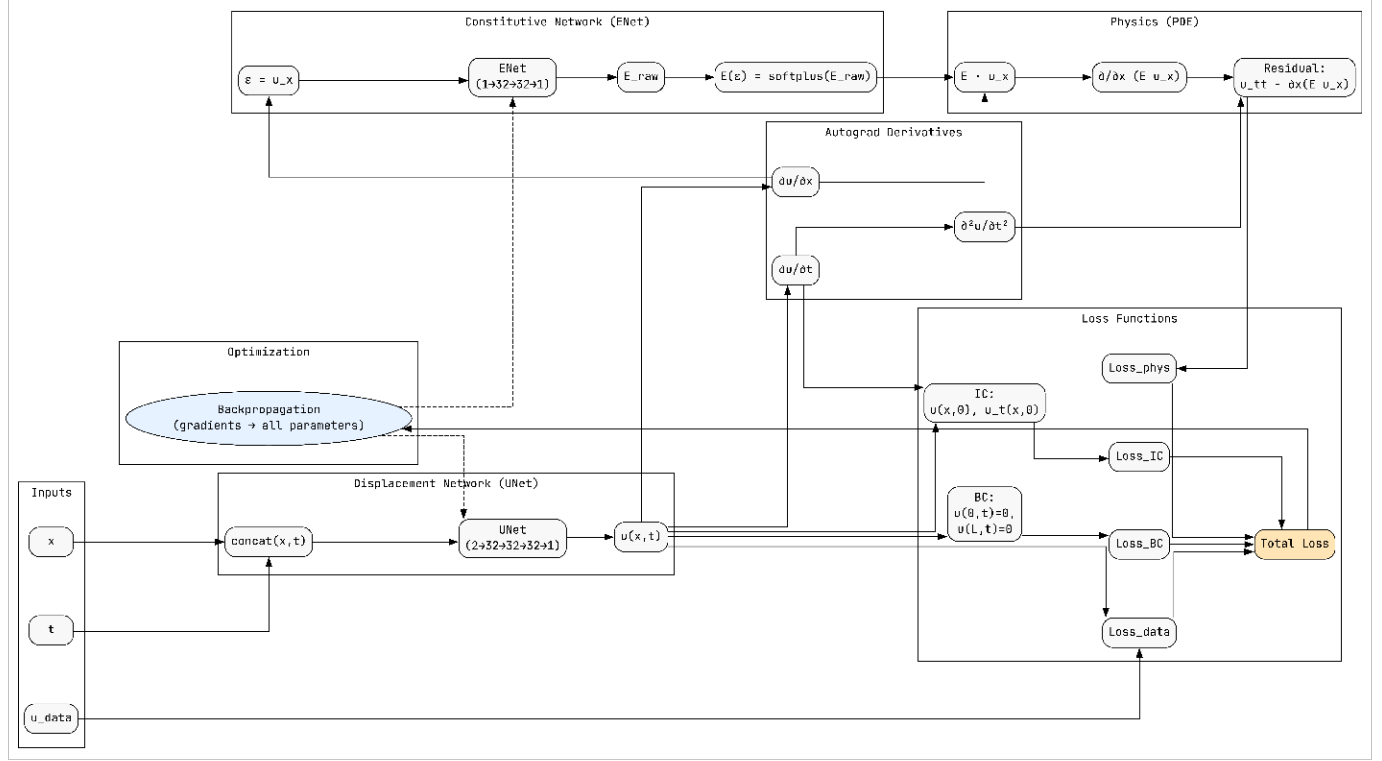


## Constitutive Relation



The PINN is unable to learn the non-linear constitutive relation and instead default to a constant Young's modulus. Further the spectral bias of the PINN is evident given that it learns the low frequencies and fails to learn the high frequencies in the solution.

## Architecture



## Question 2

The advection-diffusion equation is given by,

$$\frac{\partial u}{\partial t} + \alpha \frac{\partial u}{\partial x} - \epsilon \frac{\partial^2 u}{\partial x^2} = 0, \quad x \in \Omega, \quad t > 0$$

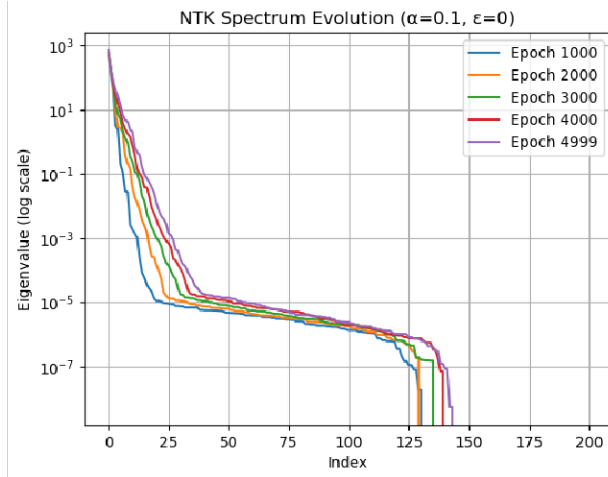
with initial and boundary conditions,

$$u(x, 0) = f(x), \quad x \in \Omega$$

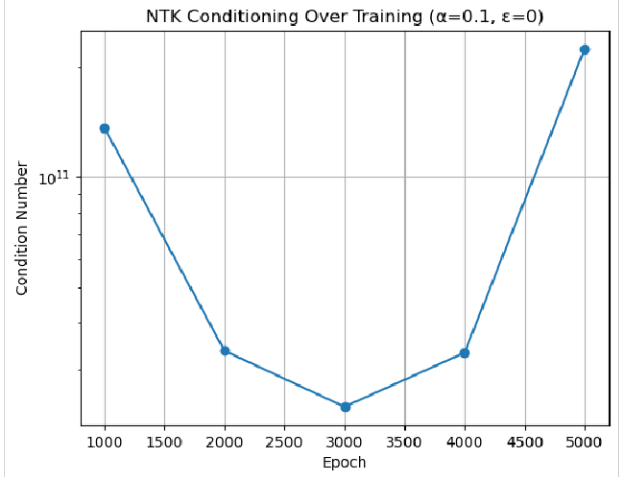
$$u(x, t) = g(x, t), \quad x \in \partial\Omega, \quad t > 0$$

## Discussion

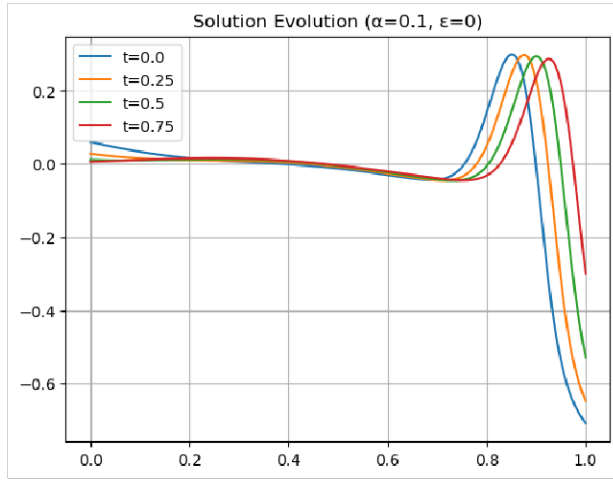
- In the Fourier spectrum and error spectrum plot, we see that there are peaks in the advection dominated regime which correspond to the dominant frequencies in the solution. In contrast, the diffusion dominated regime displays a monotonic decay in the Fourier spectrum, indicating that the solution is smoother and has less high-frequency content.
- Over epochs, the Fourier spectrum evolves unevenly in the advection dominated regime, with certain frequencies being learned faster than others. In the diffusion dominated regime, the Fourier spectrum evolves more uniformly, reflecting the smoother nature of the solution.



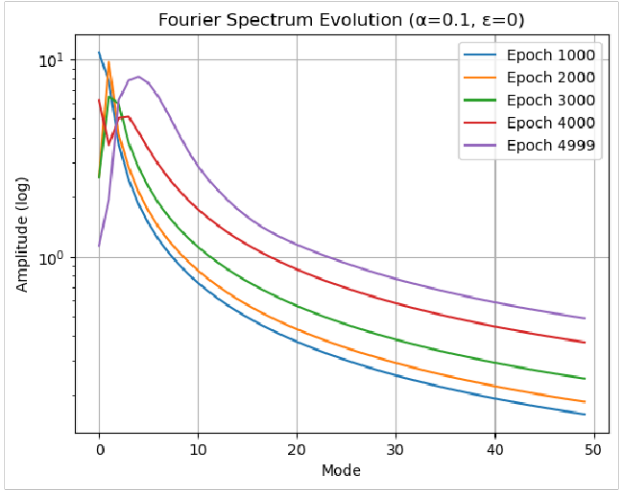
(a)



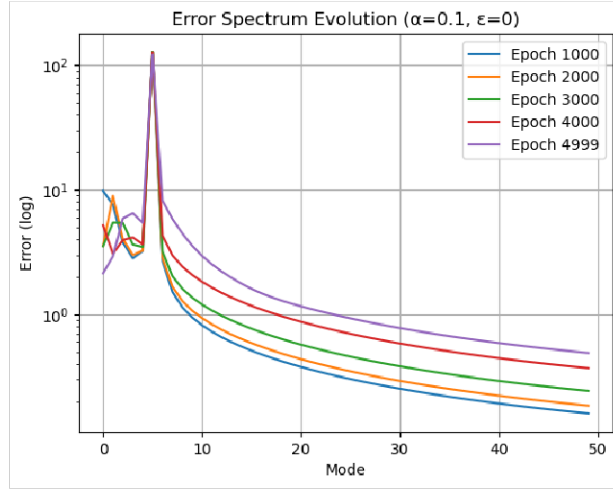
(b)



(c)

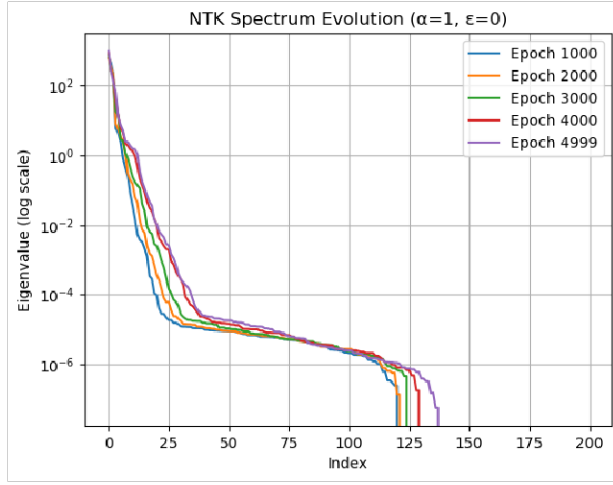


(d)

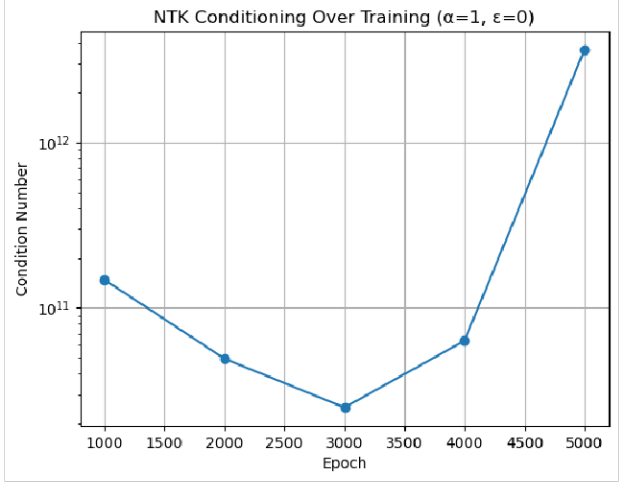


(e)

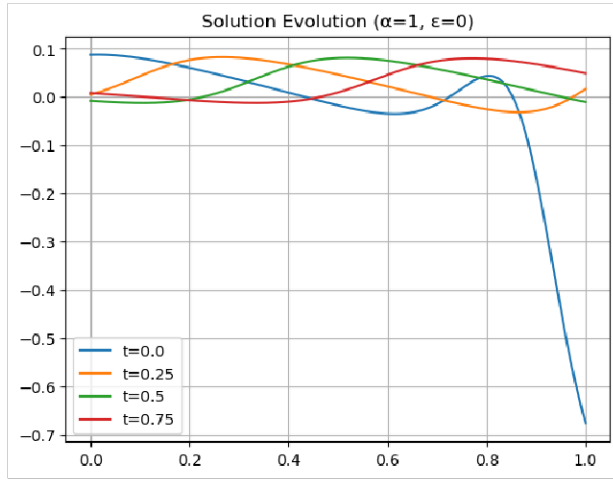
Figure 1: Advection Dominated Regime ( $\alpha = 0.1, \epsilon = 0$ )



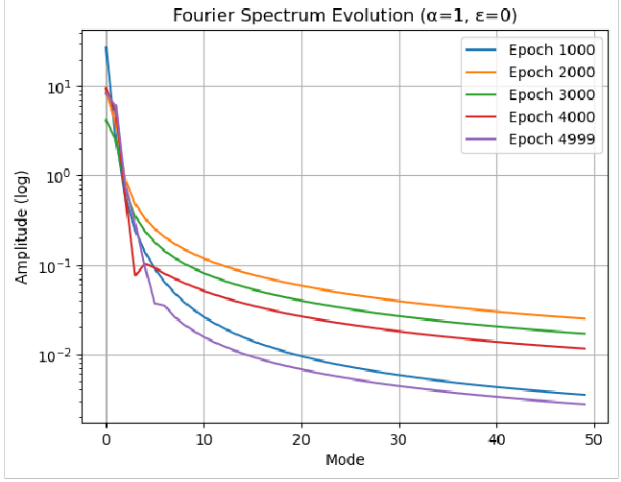
(a)



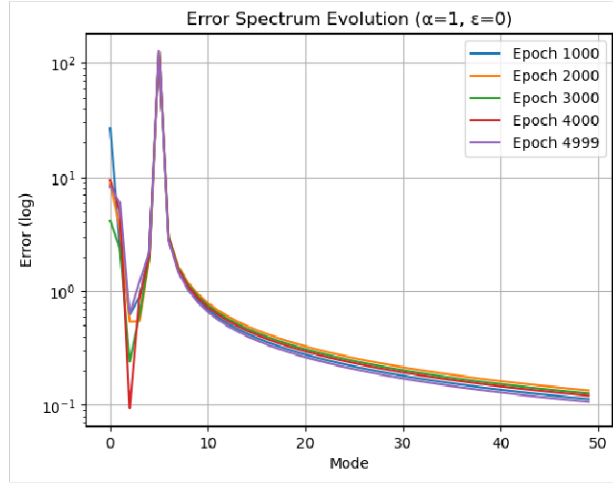
(b)



(c)

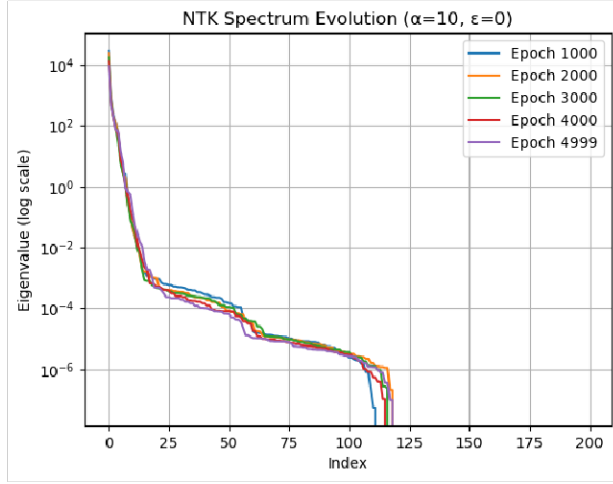


(d)

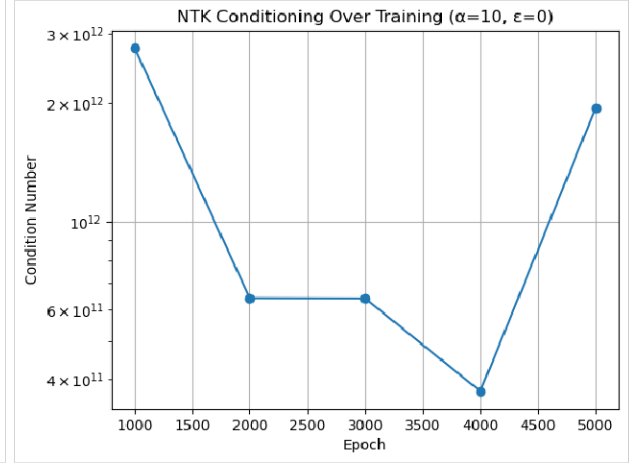


(e)

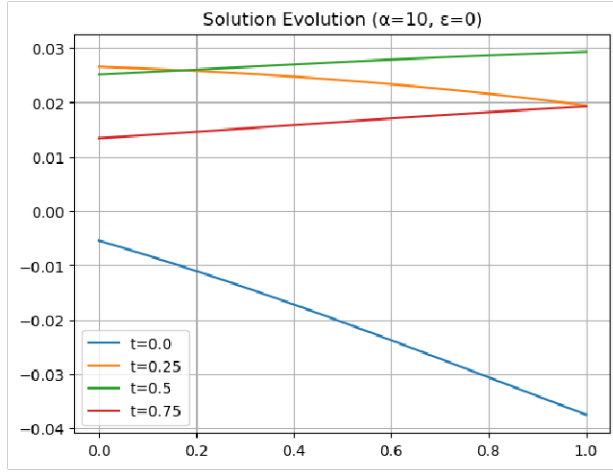
Figure 2: Advection Dominated Regime ( $\alpha = 1, \epsilon = 0$ )



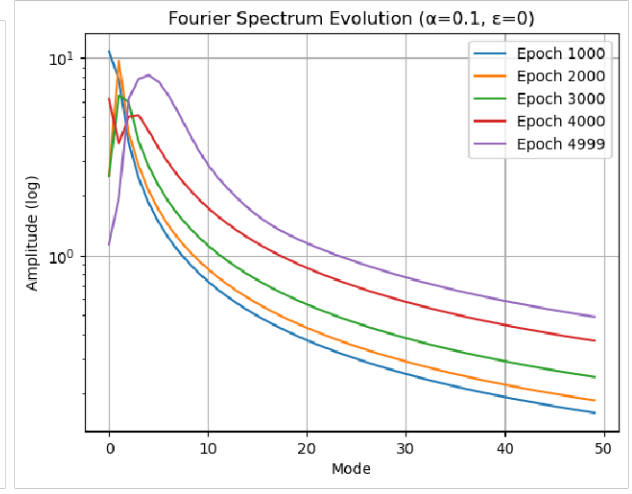
(a)



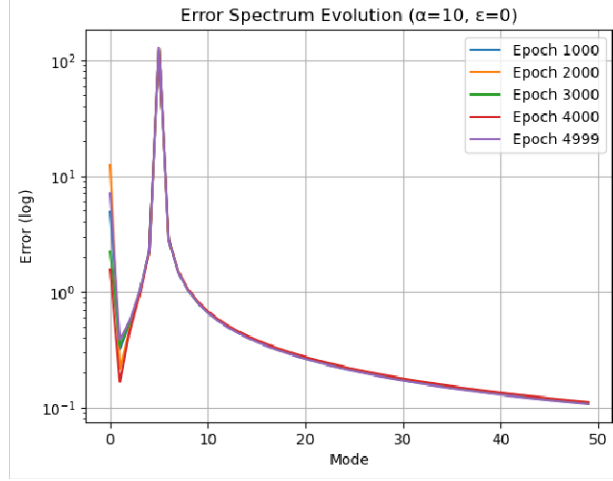
(b)



(c)



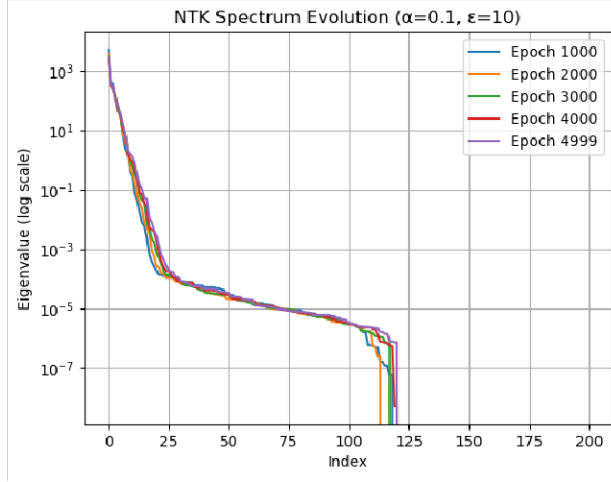
(d)



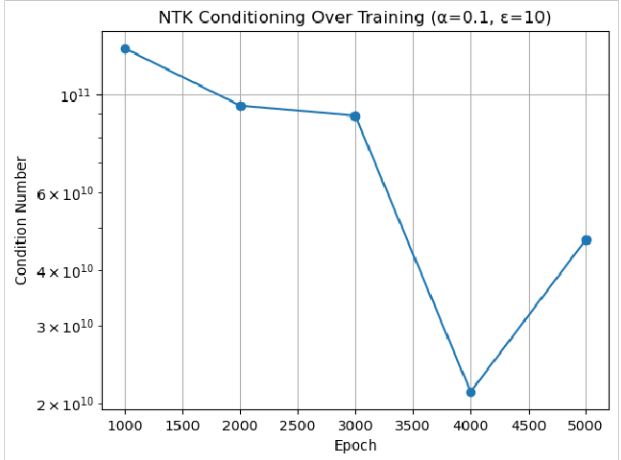
(e)

Figure 3: Advection Dominated Regime ( $\alpha = 10, \epsilon = 0$ )

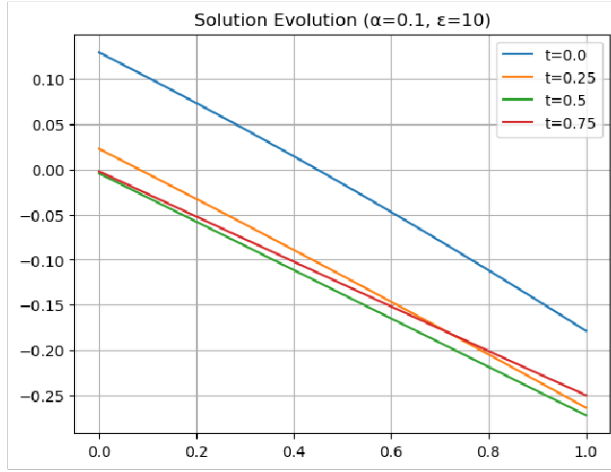




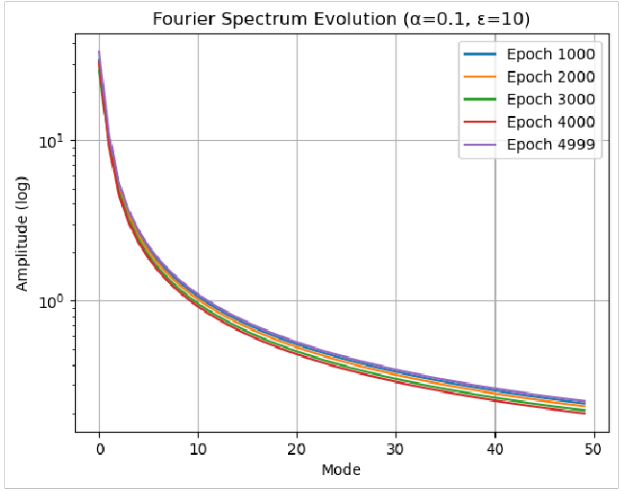
(a)



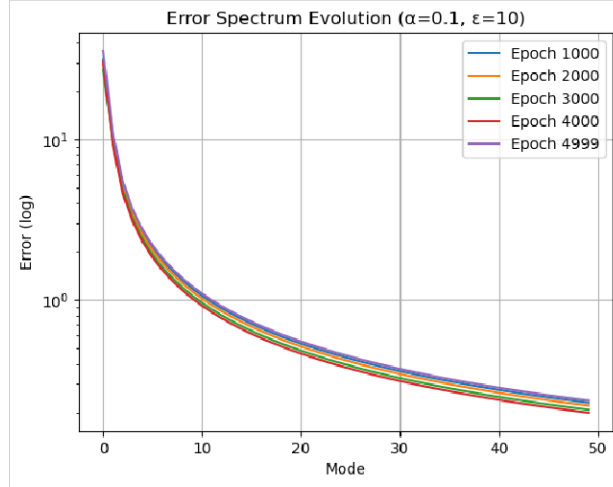
(b)



(c)

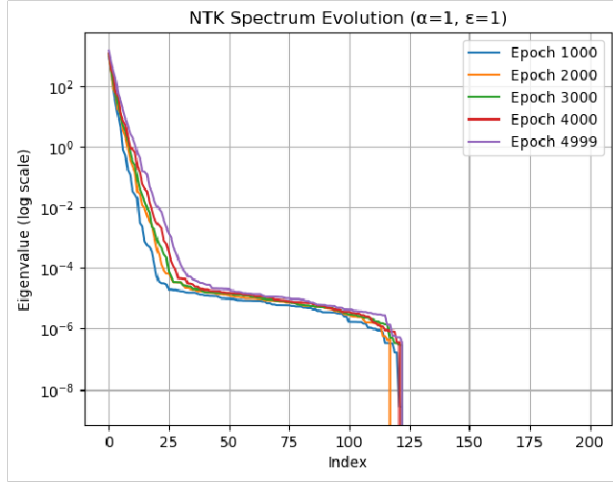


(d)

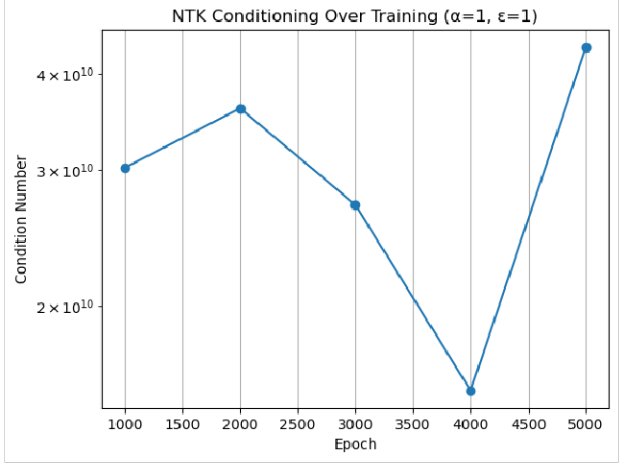


(e)

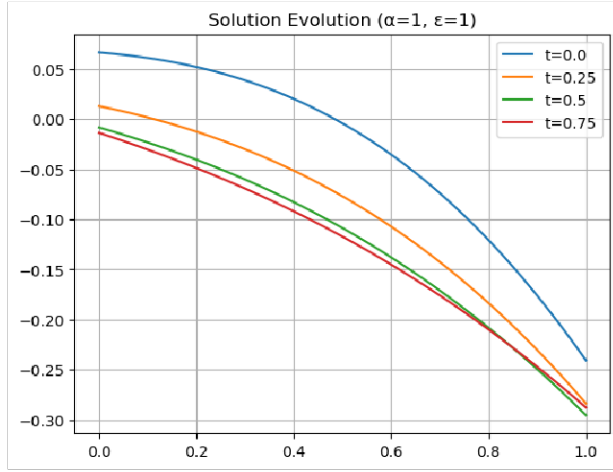
Figure 4: Diffusion Dominated Regime ( $\alpha = 0.1, \epsilon = 10$ )



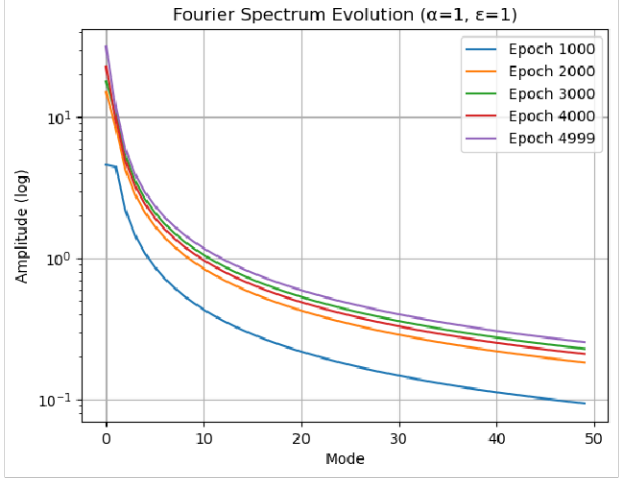
(a)



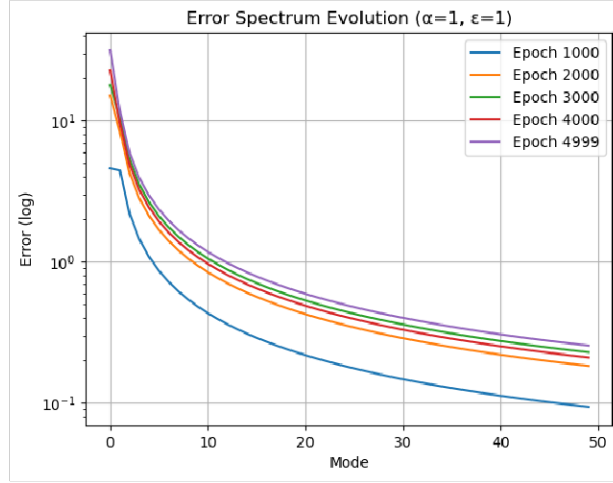
(b)



(c)



(d)



(e)

Figure 5: General Regime ( $\alpha = 1, \epsilon = 1$ )